

Task 2: Data Analytics Case Study

Instructions:

1. Choose a Field That Interests You:

- Examples: Finance, Healthcare, Sports, E-commerce, Education, social media, Agriculture, Transportation, etc.

2. Imagine a Realistic Scenario:

- What kind of organization is it?
- What kind of data might they have?
- What business problem are they facing?

3. Write 3 Business Questions they might ask, one for each type (you can mix or repeat types):

- Descriptive → What is happening?
- Diagnostic → Why did it happen?
- Predictive → What might happen next?
- Prescriptive → What should we do?

4. Classify the Analytics Type for each question.

1. Field: Sports (Football)

2. Scenario

1. What kind of Organization is it?

A football club's data analytics team.

2. What kind of Data might they have?

Player fitness levels (running distance, speed, heart rate), match performance (passes, shots, tackle success), and injury records.

3. What Business Problem are they facing?

The team's performance dropped badly, and many key players got injured, costing the club points and money.

3 & 4. Business Questions

1. How many wins and injuries did we have in the second half of the season?

Descriptive Data Analytics

2. Why were players getting injured late in matches toward the end of the season?

Diagnostic Data Analytics

3. Which players are most at risk of injury in the next few matches?

Predictive Data Analytics

4. How should we rotate players to avoid future injuries?

Prescriptive Data Analytics